

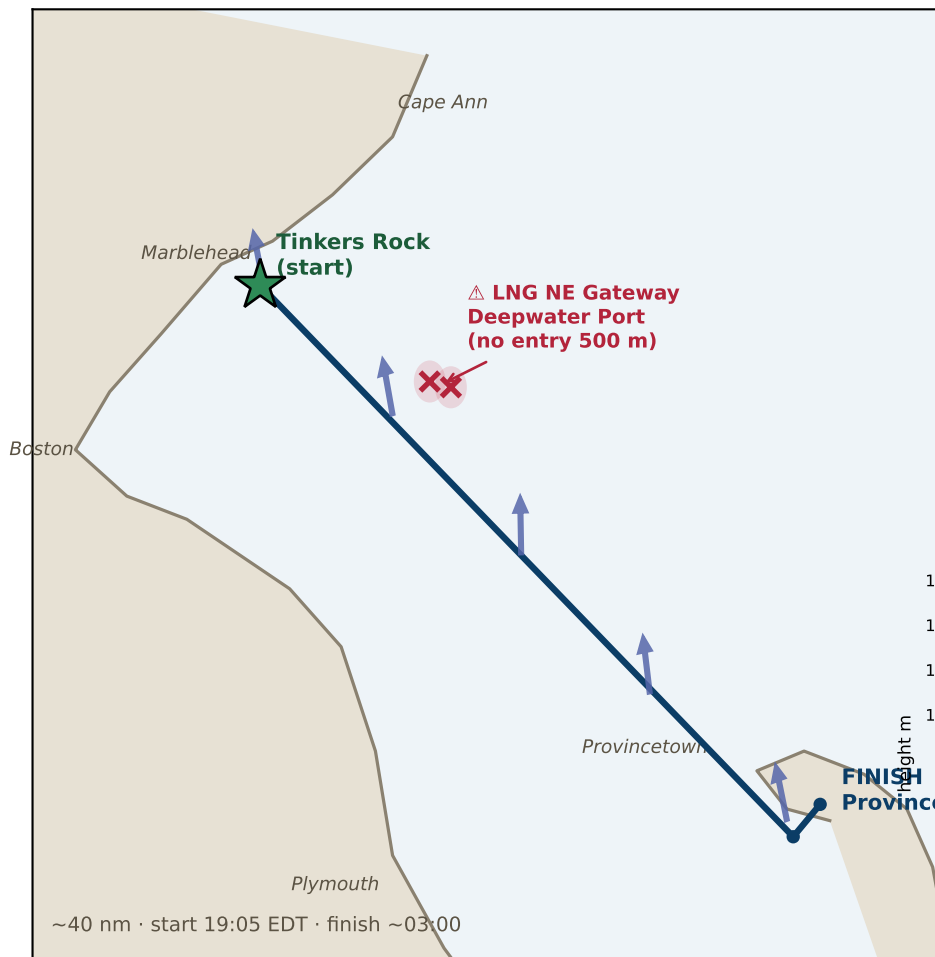
Marblehead → Provincetown — Overnight Race Weather

Friday 2026-07-17, 19:05 EDT start · ~03:00 Sat finish · ~40 nm SE across Massachusetts & Cape Cod Bays

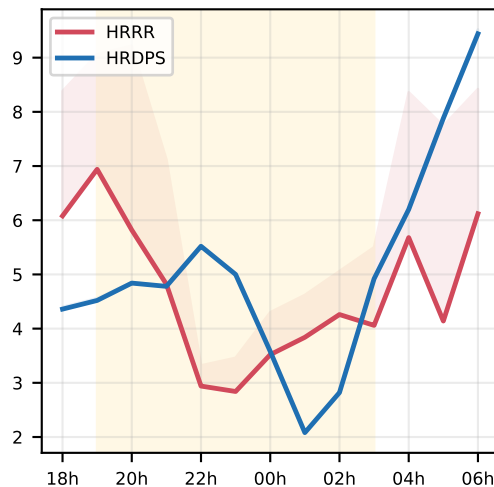
Model runs — NOAA HRRR 3 km: 2026-07-17 21Z · Env. Canada HRDPS 2.5 km: 2026-07-17 18Z (latest available cycles)

retrieved Jul 17, 18:07 EDT (2026-07-17 22:07 UTC) · ~58 min before start · models agree on the overnight, split on the start.

Route & overnight wind (HRRR mean arrows)

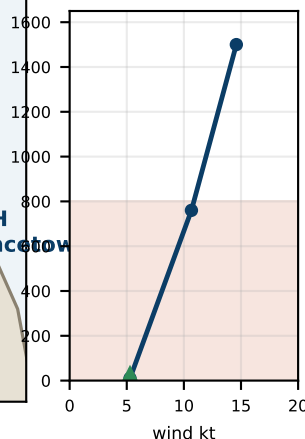


Course-mean wind (kt)

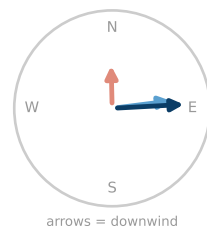


Wind shear (HRRR, mid-bay)

speed ~2x / km



veers ~100°



Overnight synopsis

Light southerly night on the EAST side of an approaching DRY trough (surface stays flat ~1016 mb — the trough and its front hold to the west; models keep the race window rain-free with no pressure fall). The models AGREE on the overnight: the wind fills S/SW a LIGHT ~7-9 kt offshore (gusts to ~13), a starboard-tack reach that gradually frees toward Race Point, seas ≤2 ft. They SHARPLY DISAGREE on the start: HRRR SE ~7 kt (a reach/beat down the rhumb) vs HRDPS a WNW land-breeze ~8 kt (offshore), dying to near-calm by ~01:00 before the southerly fills — opposite first-leg angles. Neither is reliable at a post-sunset transition: sail the breeze on the water, then get offshore into the steadier southerly. Trough watch: skies likely OVERCAST (dark, starless) and the SW FRESHENS near/after dawn (~11-12 kt, gusts ~15) — recheck today's short-lead runs for earlier timing.

Wind shear — what's happening

After sunset the sea-cooled surface decouples from the wind above — HRRR shows a low-level inversion (the air ~0.8 km up is +0.4°C WARMER than the surface). The course feels a light thermal breeze; aloft the true gradient, W-WNW 13-16 kt, flows almost freely — the two are ~100° apart and speed nearly doubles in the first km. A puff is a brief down-mix of that faster, veered air; by dawn the layer re-mixes and the surface veers to W/SW. → what to DO with it: tactics bullet ⑤ (p.2).

Conditions at a glance

Gusts — steady, modest

HRRR gust factor ~1.4-1.5, peaks ~15-16 kt over ~9 kt mean. CAPE 0 + only ~11-15 kt aloft → no convective or momentum-driven gusts. HRDPS: no gust field. Plan smooth, not puffy.

Sea state — benign

~0.3 m / 0.8 ft short-period wind chop (NWS seas ≤1-2 ft). SST ~21°C / 69°F. Flat water — the limiter is wind, not waves.

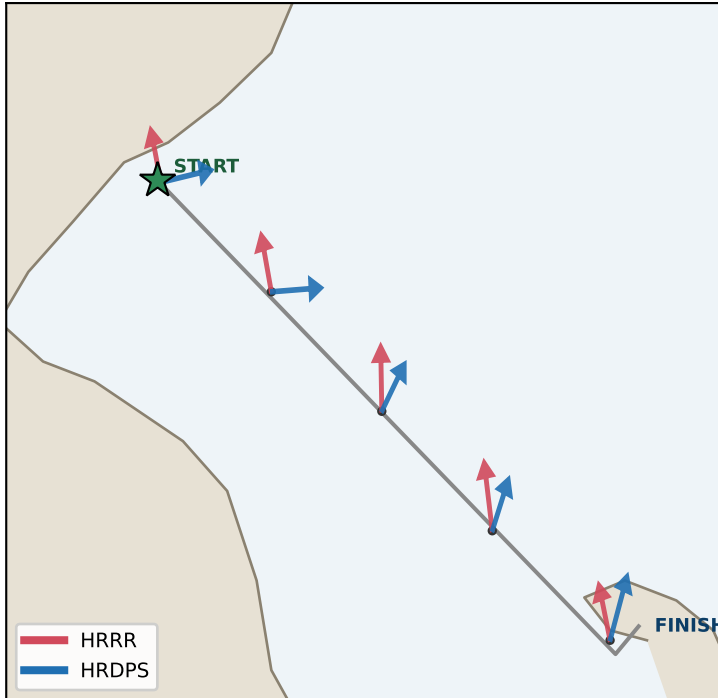
Temperature — mild & near-steady

~18°C / 64-65°F all night on the water (sea-moderated: ~1°F drop offshore, up to ~10°F near the P-town shore). Warm water under cooler air → damp; reinforces the fog watch. Layers for damp, not cold.

Tactics, Strategy & Timing

Marblehead → Provincetown · Fri 2026-07-17 overnight

Where the models differ — mean-wind arrows (length $\propto$ speed)



HRRR vs HRDPS — mean wind by point

Start (Tinkers Rk)

HRRR 165°/4kt · HRDPS 260°/6kt → Δ dir 94°

Off Nahant

HRRR 167°/4kt · HRDPS 267°/6kt → Δ dir 100°

Mid Mass Bay

HRRR 179°/5kt · HRDPS 213°/4kt → Δ dir 34°

Stellwagen edge

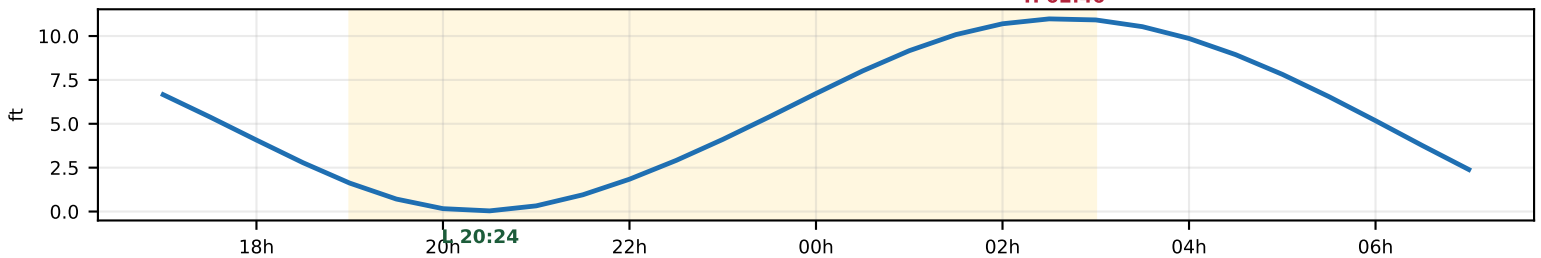
HRRR 171°/6kt · HRDPS 203°/4kt → Δ dir 32°

Race Pt / P-town

HRRR 165°/4kt · HRDPS 200°/6kt → Δ dir 35°

Biggest split at the START (near-opposite: a dying SE sea-breeze vs a NW land-breeze). They converge to a common S/SW by Race Point, HRDPS lighter throughout.
→ The first leg is the uncertain part — sail the actual breeze; for the rest of the night both models agree.

Tide at Boston (proxy) — start $\approx$ low slack, finish $\approx$ high slack



Tactics & strategy

① Start (19:05-21:00) — models disagree, so read the water

HRRR: SE ~7 kt (reach/close-hauled). HRDPS: WNW land-breeze ~8 kt (offshore), both dying after ~21:00. Opposite angles — don't pre-commit; sail the actual dock breeze (it's LIGHT, $\leq$ 8 kt). NEW: the inshore band goes near-calm (2-3 kt, shifty) ~22:00-02:00 while offshore holds 6-8 kt — so get clear air off the line and work OFFSHORE hard; the Marblehead shore is a parking lot mid-race.

② The crossing (21:00-01:00) — reach in the filling southerly

Both models fill S/SW a light ~7-9 kt offshore (gusts ~13), building to ~11-12 only near dawn — a starboard reach that frees as the wind veers right. Pressure lives offshore/south (mid-bay ~10 kt vs shore 2-6). Commit south; stay on the pressure side, don't over-stand north toward Cape Ann.

③ ⚠️ LNG Northeast Gateway — hard avoid

Two STL buoys at ~42.40 N (–70.60/–70.59) sit just N of the direct line. NO ENTRY within 500 m, no anchoring within ~1 nm. The rhumb passes ~5 nm south — staying on/south of the line is both faster (more breeze) and legal.

④ Race Point & finish (01:00-03:00) — the tide gate

S/SSW 8-11 kt, gusts to 15 into Cape Cod Bay. Flood builds to Boston high 02:45 $\approx$ finish. Round Race Point BEFORE ~02:45 to carry the flood in; later boats meet the building ebb. Finish current itself is minor (near high slack).

⑤ Wind shear — shifty, veered puffs; trim for twist

Light SE/S surface under a W-WNW 13-16 kt flow (inversion = decoupled; diagram + explainer on p.1). Puffs arrive VEERED toward W/NW and a bit stronger (cap ~16 kt): upwind, starboard lifts / port headers — tack on the headers, don't commit a side. Trim for TWIST — ease the upper leech (the masthead sees the stronger, veered wind). By dawn the surface veers/builds to W/SW.

⑥ Night, sky & hazards

Dark — thin crescent sets early + likely OVERCAST ahead of the trough (HRRR cloud, HRDPS clearer) → nav by instruments, watches + lights. Approaching DRY trough stays west during the race (no rain, flat pressure); the real SW build (12-17 kt, gusts ~18) holds off to ~05-06 AM — after most finish. FOG: not in tonight's NWS forecast (ANZ231/233 clean) — low risk. Seas $\leq$ 1-2 ft (ANZ233 gusts ~20 kt early eve), no SCA.